PERCI v2: A Scrutinised Re-analysis of the Uranus-Moon Penetrator Secondary Payload

Independent audit, probabilistic delivery–impact–thermal–power–link models, a novel-methods study, and a proposed v2 design

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Abstract

The Penetrator for Environmental, Regolith, and Crater Investigation (PERCI) was proposed in a 2025 concept study as a ∼119 kg penetrator released from the Uranus Orbiter and Probe (UOP) flagship 4.4 h before closest approach to Umbriel. In that design the penetrator is retargeted by four STAR 5D solid motors (a 450 m/s requirement), impacts at 0.9 km/s and survives 30 min on a 140 W heater. In this paper every equation, table and budget of the study was audited, which produced 61 findings (4 critical, 21 high). We then rebuilt the delivery, penetration, thermal, power and link models as deterministic, tested, probabilistic codes and ran a novel-methods study. The report's Umbriel trajectory row is reproduced within 4% in impact speed, but the minimum-ΔV central hit needs only 323–328 m/s and arrives at 0.75–0.76 km/s. The 140 W heater was found to be an artefact of a battery-only thermal mass and a 1 W/K mount assumption: with an insulated core of ordinary heat capacity, 30 min needs 0 W and 2 h needs $\leq$24 W in 90% of sampled cases. The link as written does not close, because the flagship carries no S-band; at UHF the median 10 kbps margin is -7.5 dB at 5.4–6.5 Mm, and of the planned 20 GB of data only about 16 MB can be returned. No impact model is validated above 341 m/s in ice, and the report's 3.4–8.3 m depth is the stiff-ice corner of a distribution spanning 2–600 m. Finally, the wheel sizing for the 10°/s flip is 43–66× too small, but the flip itself is unnecessary because the minimum-ΔV burn is 67°, not 180°, from the impact velocity. A v2 baseline is proposed together with a list of what remains unknown. The penetration physics at 0.75 km/s is at TRL 2–3 and is the dominant open risk.

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Nomenclature

Trajectory and Delivery Parameters	
ΔV	retarget velocity change [m/s]
v_{imp}	impact speed [km/s]
T	release time before closest approach [h]
$g = h/(R\,v)$	signed aim parameter (h impact parameter, R moon radius, v relative speed)
P(acceptable)	probability of an acceptable impact (criteria in Section 4)
Penetration Parameters	
F	resisting force on the penetrator [N]
z	penetration depth [m]
a, ψ	penetrator radius [m] and nose caliber-radius-head
N, f_{nose}	nose shape factor and nose factor
R_t, Y, E, ρ_t	target resistance, yield strength, Young's modulus [Pa] and density [kg/m^3]
m, α	penetrator mass [kg] and incidence angle
Thermal Parameters	
C, C_s	core and shell heat capacity [kJ/K]
G, G_{ins}	conductance to the ice and core-to-shell conductance [W/K]
P	heater power [W]
k, c	thermal conductivity [W/m/K] and specific heat of the target
Risk Parameters	
S, P, D	severity, probability and detectability scores
RPN	risk priority number

1 Introduction

The Uranus Orbiter and Probe (UOP) flagship [1] would tour the Uranian system and make several ballistic flybys of the major moons. These flybys only allow remote observations, so in-situ measurements of surface strength and thermal properties remain out of reach of the orbiter alone. The 2025 PERCI concept study [2] proposed to fill this gap with a penetrator released from the flagship during a flyby and delivered to the surface of Umbriel.

However, a concept of this kind rests on a chain of models (delivery, impact, thermal, power and link), and an error in any one of them changes what the mission can return. The headline numbers of the 2025 study had not been independently checked, and several of its budgets disagree with each other. The goal of this paper is to address this gap. This study audits the 2025 report, rebuilds each of its models as deterministic, tested code with stated uncertainty, tests a set of new delivery and design ideas against the same baseline, and proposes a v2 design.

The rest of the paper is structured as follows. Section 2 describes the approach, the inputs and the literature used. Section 3 presents the audit of the 2025 report, and Section 4 the revised requirements and concept of operations. Section 5 gives the results of the delivery, impact, thermal, power, link and science models, together with the novel-methods study. Section 6 discusses the recomputed risk register, the v2 design, the verification plan, the sources of error and model limitations, and the objections a reviewer would raise. The paper concludes with an assessment of the concept and the work that should follow in Section 7.

1.1 Summary of Findings

Verdict. The concept is not invalidated by this re-analysis, but its headline numbers are not defensible as published. Four findings are critical: the link budget (B-01, B-02), the reaction-wheel sizing for the SRB flip (B-28) and the thermal 140 W sizing (A-01). The study also has real strengths. Its Umbriel trajectory row is independently reproduced, its power-budget arithmetic is internally consistent, its STAR 5D data match the manufacturer's catalogue, and its risk-matrix arithmetic is consistent (the "OK" rows of Table 8).

Corrected facts. The corrections that change a reported value are as follows:

- **Impact.** The summary and the appendix give different impact numbers (6.5–16 MN and 2.7–3.5 m versus 5.4–10.4 MN and 3.4–8.3 m). The Tate–Alekseevski input R_t is 0.454 GPa by the report's own formula, not 0.6 GPa, and the Glen's-law "flow stress" is a creep law applied far outside its range (A-03, A-04).
- **Link.** The table's 3.3 bps "Shannon limit" comes from a unit slip (0.125 Hz used for 0.125 MHz); the correct figure is 842 kbps. The EIRP is 11.4 dBW, not 5.67 dBW, and the receive gain is 24.2 dBi, not 15.0 dBi. The claims of "15 dB at 10 kbps" and "1 dB at 2.14 Mbps" cannot both hold (B-02..B-05).

- **Power.** The 37 W average is not reproducible (we obtain 29.2 W in descent and 143.8 W post-impact), whereas the 705,145.5 J total is reproducible (29.175 W×4.25 h + 143.76 W×0.5 h) (B-13, B-14).
- **Mass.** The 159 kg figure is a *wet* mass compared against a *dry* 180 kg limit. With AIAA S-120 growth the wet mass is 191 kg (dry 163 kg) (B-33).
- ΔV**.** With the report's own masses the 4-motor capability is 483 m/s, not 476 m/s, and the margin disappears with 9% mass growth (B-22, B-23).

v2 design changes. The changes are detailed in Table 6. The ΔV requirement is lowered from 450 to 330 m/s, with a 22% capability margin at MEV mass, and the impact speed falls from 0.90 to 0.75 km/s. The 180° flip is removed and replaced by a slow 67° slew. The heater power drops from 140 W to 0 W for 30 min and to ≤24 W (90th percentile) for 2 h. The drag flare is replaced by a split forebody/aftbody with a surface antenna, and the link becomes a UHF 6 kbps link with a corrected budget. The science is descoped from interior structure and plume composition to near-surface strength, thermal conductivity and descent imagery.

Novel-methods workstream. In this workstream (Section 5.7) 14 ideas were simulated: 4 are kept, 5 are partial, 1 is unproven and 4 failed. The strongest numerical result is that release times of 12–24 h cut the retarget ΔV by a factor of 3–6 and the propulsion mass from 26 kg to 3–9 kg. However, P(acceptable impact) then depends on the unknown flyby orientation (0.47–1.00 at 24 h). We are strict about what is new: only the UOP/Umbriel-specific numbers and the robustness analysis. The flyby-released penetrator itself has prior art (Wurz et al., 27 h before closest approach).

Remaining unknowns. The main unknowns (Section 6.4) are the penetration of 85 K ice/regolith at 0.75 km/s; the shock survival of a broadband seismometer and of the cells above 5,000 g; the UOP UHF antenna gain, G/T and tour state; the Umbriel regolith thickness; and the SRB behaviour after 13.4 years of cold soak.

2 Approach

The goal of this study is to test whether the 2025 PERCI concept closes once each of its models is recomputed independently, and to propose a design that does. The analysis links

1. the delivery trajectory from the flagship flyby to the moon,
2. the impact and penetration of the target,
3. the thermal, power and link budgets, which set how long the penetrator survives and how much data it returns, and
4. the science return that follows from these,

and every result is checked against the corresponding number in the 2025 report.

2.1 Inputs and Provenance

The inputs are the 85-page final report [2], the UOP study [1], and a verified literature and data survey kept in `research/`. The survey covers 53 sources, of which 38 were opened in full and 15 are snippet-only, and each parameter in it is tagged VERIFIED, DERIVED or UNVERIFIED.

2.2 Method

Every number in this report is produced by deterministic code (`analysis/`, driven by a single `run_all.py` and checked by 52 unit tests), and every random draw is seeded. Where a parameter is not in the literature it is tagged ASSUMED in the code, and the result is reported as a band over the assumption, never as a point. Spreads are taken over three sources: the four unknown orientations of the UOP flyby geometry, Sobol-sampled target and material properties, and a fixed-seed Monte Carlo of the error budget.

2.3 Replicated and New Elements

Several parts of the analysis are replicated, standard methods: circular-restricted three-body shooting (Hill/Clohessy–Wiltshire dynamics [3]), cavity-expansion penetration (the Forrestal–Luk family [4, 5]), lumped-capacitance and radial conduction thermal models, link-budget bookkeeping and FMEA arithmetic. The elements that are new in this work are the quantified UOP/Umbriel delivery families and their robustness to the unknown

flyby orientation (Section 5.1), the reverse-engineering of the report's thermal table (Section 5.3), the combined probabilistic acceptance model, and the model-referenced risk register. We make no claim of novelty beyond those and the audit itself.

2.4 Honesty Rules

No source or datum is invented in this paper. Items that could not be verified are labelled UNVERIFIED, for example the ice RF loss at 100 K and 2 GHz, the Forrestal–Luk constants for ice, whether UOP carries a spectrometer such as L'Ralph, the aerogel/MLI values, and the Uranian-moon ephemeris error in 2047.

2.5 Literature and Data

The key verified inputs are listed here; the full tables are in `research/params.json`. For Umbriel, $GM = 85.1 \pm 1.9$ km^3/s^2 and $R = 584.7$ km [6], the escape speed is 0.540 km/s, and the orbit has $a = 266{,}000$ km and $P = 4.144$ d [7]. For Uranus, $GM = 5.794 \times 10^6$ km^3/s^2 [8]. The UOP Umbriel flybys are on 22 Jun 2047 (5,354 km, 0.52 km/s) and 3 Oct 2047 (3,653 km, 0.73 km/s) [1]. The UOP study lists X-band, Ka-band and a single UHF low-gain antenna with a 6 kbps probe relay, and it includes no penetrator [1]. Umbriel's thermal inertia is 5–15 SI units, against 2000–2600 for solid ice [9], so the surface is porous regolith whose thickness is unknown [10].

Several assumptions in the report are contradicted by the literature (the full list is in `research/contradictions.md`):

1. An impact at 0.9 km/s into ice is 2.6× the fastest documented full-scale ice impact (20 kg at 341 m/s, 24,000 g [11]). Young's empirical equations are valid only below 1,219 m/s and carry no data in soft materials at very high speed [12]. Flown penetrators landed at 80–200 m/s (Mars 96, DS-2) [13, 14], and none has operated [15].
2. Crary waves need a floating ice shell above an ocean [16], and none is established for Umbriel.
3. The UOP orbiter has no S-band, and its UHF antenna is a probe-relay antenna [1].
4. An impact seismic efficiency of 10^{-4}–10^{-3} is optimistic for porous regolith: Apollo booster impacts into unbonded soil gave 10^{-6}–10^{-5}, and InSight adopted 5×10^{-4} [17].
5. For penetrator instruments, the Lunar-A seismometer survived 8,000–10,000 g [18], unprotected silicon MEMS suspensions fail at tens of g [19], and the DS-2 values are 30,000 g for the forebody and 80,000 g for the cells [14, 20].
6. The DS-2 and Mars 96 designs separate the forebody and aftbody so that the antenna remains at the surface [13, 14]; the report's single body with a drag flare has no heritage.

3 Audit of the 2025 Report

Every equation, table, budget, link budget, impact model, thermal result, ΔV, mass budget and timeline in the report was recomputed independently. The scripts are `audit/recompute_*.py`, and the paper-versus-recomputed output is kept in `audit/paper_vs_recomputed.txt`. Figure 1 gives the counts by severity, and the full list is given in Table 8 (Appendix A).

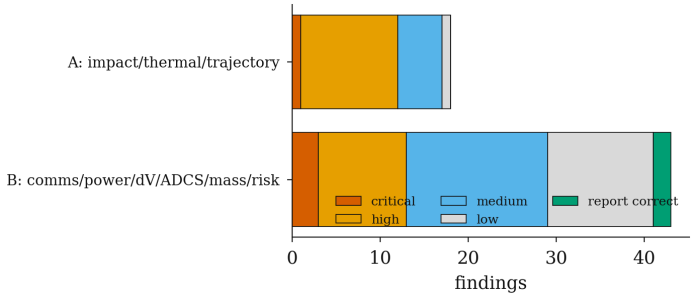


Figure 1: Audit findings by severity (61 total). Two of the findings (B-13 power sums, B-37 RPN arithmetic) confirm that the report is right.

Notes already known to the team, confirmed or refuted. Several issues were known to the original team before the audit; each was checked as follows.

1. The summary and appendix impact ranges disagree: *confirmed* (A-08).

2. The 37 W versus 29.2 W discrepancy: *confirmed*, and the 37 W has no reproducible origin (B-14).
3. T-4.6/T-4.4 h versus Table 2: *confirmed*; five different pre-impact times appear in the report (B-15, B-40).
4. The link table shows "Above sensitivity? NO" and "Shannon limit? NO" with 3.3 bps, while the text claims a 15 dB margin at 10 kbps: *confirmed and explained*. A bandwidth of B= 0.125 Hz was used for 0.125 MHz, which flips C/N by 60 dB, and the 15 dB claim reproduces only with the Rx gain corrected to 24.2 dBi and the wrong EIRP kept (B-02, B-05).
5. The thermal case labels: the 1-node "best case" is case 3 (gold polyimide, $e^* = 0.005$, 7.9 W), not case 5, and it refers to a 1.6×1.06×0.96 m box, not to the penetrator (A-02).

New critical/high findings. The audit also produced new findings. Table 34's heater powers are reproduced by a one-node battery-only model (Section 5.3). The report's flare would raise the peak deceleration 4–10× (Section 5.2). Finally, a Titania target is reachable, which contradicts the "820 m/s" row, because the report used the 2045 fast flyby rather than the 0.47 km/s flyby shown in its own Fig. 14 (A-13).

4 Requirements and Concept of Operations (v2)

Inherited, unchanged: the requirements carried over from the 2025 study are COTS/TRL 6+ components, the SIMPLEx guidelines, $\leq$ 2 W drawn from the orbiter in cruise, do no harm to the orbiter, UHF communication to the flagship, and a mass of $\leq$ 180 kg. **Revised (v2):** the requirements changed for v2 are

- SYS_SD_3 (ΔV capability): $\geq$ 330 m/s is required at MEV mass (it was 450 m/s); the capability is 400 m/s with four STAR 5D motors at 149 kg (Section 5.1).
- New: P(acceptable impact) ≥ 0.9 for every plausible flyby orientation. An impact is *acceptable* when it is a hit, the incidence is $\leq 10°$, the impact speed is ≤ 0.95 km/s, the flagship is $\geq 5°$ above the local horizon at impact+30 min, and the slant range is $\leq$ 7,000 km. These criteria are ASSUMED and traceable to the structural model.
- SYS_SD_1 (survive 30 min): this is now demonstrated with a 0 W heater. A 2 h hold at $\leq$ 20 W is met for $\sim$ 88% of the sampled thermal parameter space, and 6 h is kept as the stretch objective.
- Link: the rate is capped at 6 kbps (the UOP probe relay [1]) unless the UOP UHF G/T is confirmed to be higher. The memory is sized to the link (1 GB class), not to 50 GB.
- Decadal traceability: Q5.1c (deceleration/strength profile), Q8.2f (bulk thermal conductivity) and Q12.8a (partial: descent imagery) are kept, while Q5.1a (interior structure) and Q4.4b (plume composition) are descoped (Section 5.6).

CONOPS (v2). The v2 concept of operations runs as follows. The penetrator is released at T−4.4 h and spun up, and it burns along the ΔV vector with the SRB nozzles aft and the nose forward. It is then despun and performs a slow ($\leq 0.15°$/s) slew of ≈67° to the velocity-tracking attitude, which is completed in ≈450 s on 0.06 N m s wheels. A coast of 4.6 h follows, with UHF telemetry at the rate the link supports, and the core is preheated during the last hour. Impact occurs ≈12 min after the flagship's closest approach. The accelerometer records the impact pulse, the RTD decay and the seismometer are recorded for ≥30 min, and the split aftbody carries the antenna.

5 Results

This section applies the models described in Section 2 to the delivery, impact, thermal, power, link and science problems in turn, and then tests the novel design ideas against the same baseline.

5.1 Trajectory, Release and Delivery

5.1.1 Model

The delivery is modelled with planar circular-restricted three-body dynamics (a Uranus point mass and an Umbriel point mass on a circular orbit, with the indirect term included), integrated with an 8th-order Dormand–Prince scheme (rtol 10^{-11}). The flagship state is fixed by the UOP flyby table (altitude and relative speed at closest approach [1]). The table does not give the orientation of the flyby, so all four combinations of (flagship outside/inside the moon's orbit radius) × (prograde/retrograde relative velocity) are run and reported as a spread. We treat the burn as instantaneous, since a STAR 5D burns for 3.03 s [21]; the finite-burn, thrust-offset and spin effects are carried in the dispersion model. Central hits (zero impact parameter, incidence $< 2°$) are found by shooting on the signed aim parameter $g = h/(R\,v)$, and minimum-ΔV solutions

by projected Gauss–Newton from several deterministic starts. Surface co-rotation (10 m/s) is included. The Uranus tide also matters: at 8,000 km it displaces the trajectory by ∼10%.

Validation against the report. For the Umbriel row of Table 2 (release at 4.4 h, $\Delta V = 415$ m/s), the independent central-hit solution gives 401 m/s → 0.86 km/s and a flight time of 3.76 h, against the report's 901 m/s, 4.2 h and 86°. This is within 4% in speed and 10% in time. The Ariel, Oberon and Titania rows are not reproduced, because the objective used to generate them is not stated (A-12).

5.1.2 Minimum ΔV versus Release Time

Figure 2 shows the minimum retarget ΔV and the matching impact speed as functions of the release time.

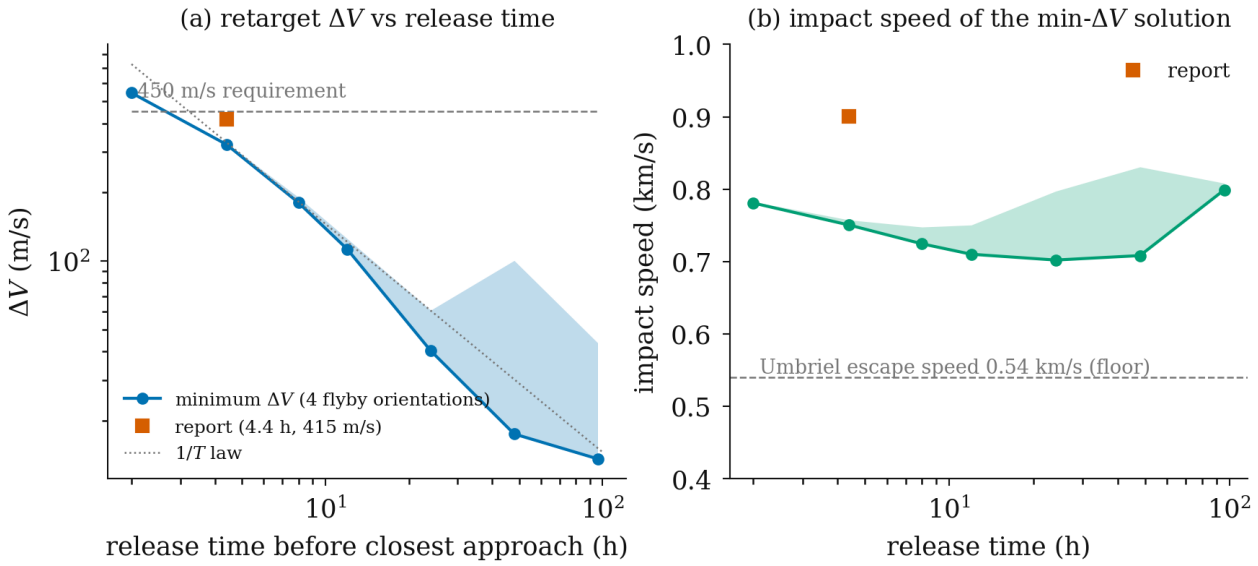


Figure 2: (a) Minimum retarget ΔV for a central hit on Umbriel versus release time; the band covers the four unknown flyby orientations. Below ∼12 h the law is $\propto 1/T$ (a straight-line retarget of a ∼5,200 km miss), while beyond $\sim 1/n = 16$ h orbital dynamics flatten it and the multi-start solver becomes unreliable, so the 48–96 h values should be treated as order-of-magnitude. (b) Impact speed of the minimum-ΔV solution, which is 0.70–0.83 km/s; the floor is Umbriel's escape speed of 0.54 km/s.

At the report's release time the minimum is 323–328 m/s (not 415, and not 450), with $v_{\mathrm{imp}} = 0.75$–0.76 km/s. Releasing 12 h earlier cuts the requirement to 121–125 m/s, and releasing 24 h earlier cuts it to 40–61 m/s (57–73 m/s for line-of-sight-feasible solutions). A burn at closest approach is the *worst* choice (855 m/s at 6 min before closest approach). The reason is that the Oberth factor available at this distance is only 1.05, because Umbriel's escape speed at 5,939 km (0.17 km/s) is far below the 0.52 km/s flyby speed.

5.1.3 Impact Speed, Line of Sight and the Pareto Front

Figure 3 shows all central-hit solutions in the impact-speed versus ΔV plane, together with the encounter geometry of two designs. With the line-of-sight constraint, the lowest impact speed reachable with $\Delta V \leq$ 300/350/400 m/s is 0.74/0.71/0.67 km/s (impact 0.15–1.5 h after closest approach); without it, the values are 0.62/0.59/0.57. The constraint therefore costs 0.1 km/s, because the flagship is leaving and the link range grows.

5.1.4 Dispersion and Probability of Acceptable Impact

Figure 4 shows how the probability of an acceptable impact falls with each error source. The error budget (1σ, ASSUMED) is: burn magnitude 0.7% (SRB) or 0.5% (thruster); pointing 0.6° per axis (the thrust offset after spin stabilisation, B-30); release timing 5 s; relative-navigation position 20 km (Uranian-moon ephemeris plus flagship OD); and separation velocity 1 cm/s. The nonlinear encounter is linearised by finite differences and checked against direct integration, and the Monte Carlo uses $N = 40{,}000$ seeded samples. For the re-simulated report concept P(acceptable) is 0.89, and for the minimum-ΔV SRB design it is 0.94, in all four flyby orientations. The dominant terms are the burn magnitude (for which ∂g is largest) and the pointing. With 1° pointing P falls to 0.75, and with 2° to 0.38; with 50/100 km navigation error it falls

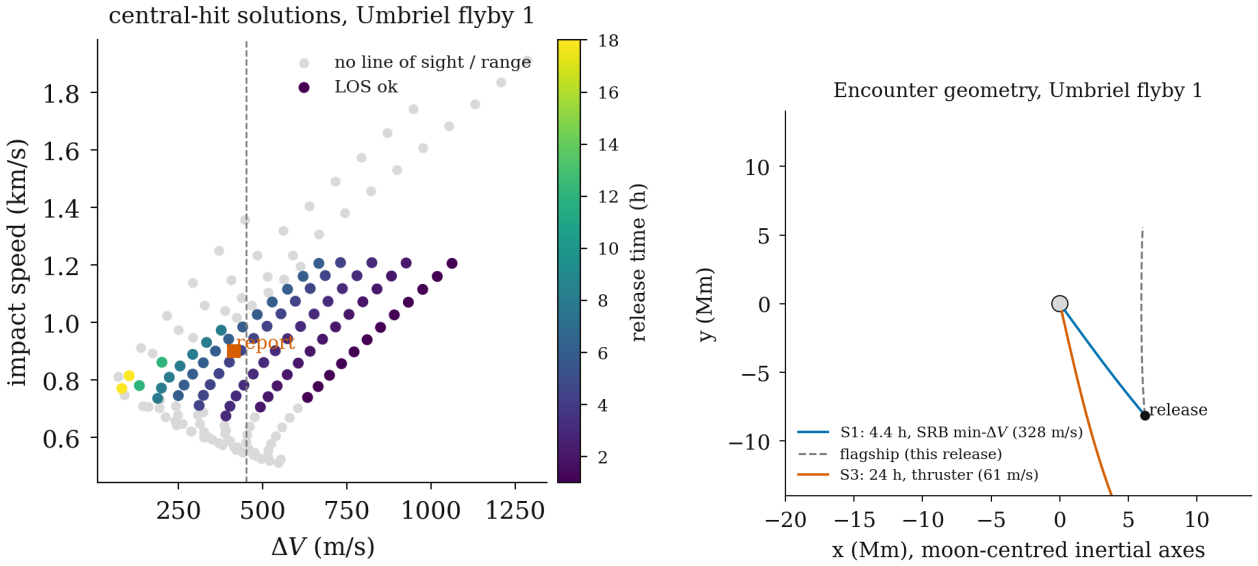


Figure 3: Left: all central-hit solutions (release 1–24 h) for Umbriel flyby 1, impact speed versus ΔV; the dark dots keep the flagship $\geq 5^\circ$ above the horizon and within 7,000 km at impact+30 min. Right: encounter geometry for the 4.4 h SRB and 24 h thruster designs.

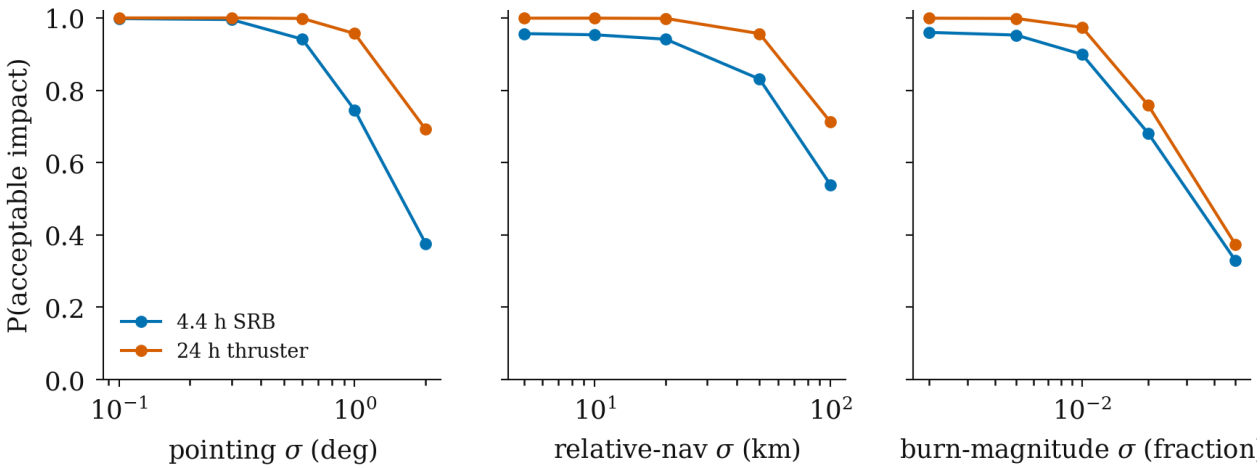


Figure 4: P(acceptable impact) versus pointing, relative-navigation and burn-magnitude error for the 4.4 h SRB design (S1) and a 24 h thruster design (S3), flyby orientation (+,+).

to 0.83/0.54, and with 2% magnitude error to 0.68. These results define requirements that the report lacks: pointing $\leq 0.6^\circ$ (1σ) including the thrust-vector offset, navigation ≤ 20 km, and impulse repeatability $\leq 1\%$.

Line of sight. In 4 of 4 orientations the 4.4 h design keeps the flagship $\geq 5^\circ$ above the horizon with probability ≥ 0.99. The earlier-release designs do not: P(line of sight) is 0.75–0.87 at 12 h, 0.58–1.0 at 24 h and 0.0–1.0 at 48 h. The report's claim of line of sight "because flight time < time before closest approach" is not a valid argument (B-11).

5.2 Impact, Penetration and Structure

5.2.1 Model

The penetration is modelled with dynamic spherical cavity expansion for an ogive nose of caliber-radius-head ψ and radius a:

$$F(z,v) = f_{\text{nose}}(z)\,\pi a^2\left[R_t(z) + N\rho_t(z)\,v^2\right] + F_{\text{friction}}, \qquad N = \frac{8\psi - 1}{24\psi^2}, \qquad R_t = \tfrac{2}{3}Y\left[1 + \ln\frac{E}{3Y}\right], \quad (1)$$

where f_{nose} is the nose factor, R_t the target resistance, N the nose shape factor, ρ_t the target density, Y the target yield strength and E its Young's modulus. The force is integrated as $m\,dv/dt = -F$. For the

report's CRH-5 nose $N = 0.065$, whereas the report's Tate–Alekseevski formula uses a blunt face ($N = 0.5$) and so over-predicts the inertial term 7.7×. We reproduce the report's 10.4 MN only with $R_t = 0.6$ GPa (the 0.454 GPa from its own eq. B.5.3.4 gives 8.8 MN), and we reproduce its second model's 5.3 MN. Three target classes are used, with ASSUMED prior weights and results shown per class: *hard ice* (Y = 20–150 MPa log-uniform, rate enhancement 1–2.5, E = 9–12 GPa), *regolith over ice* (0.5–15 m of compacted regolith, R = 1.5–24 MPa, over hard ice), and *porous regolith* (R = 0.2–8 MPa, ρ = 400–900). The penetrator mass is 110–150 kg (CBE to MEV), $a = 0.058$ m and the friction is 0.02–0.2; the inputs are Sobol-sampled with $N = 2048$.

5.2.2 Results

Figure 5 shows the deceleration pulses and the depth distributions for the three target classes. At 740 m/s

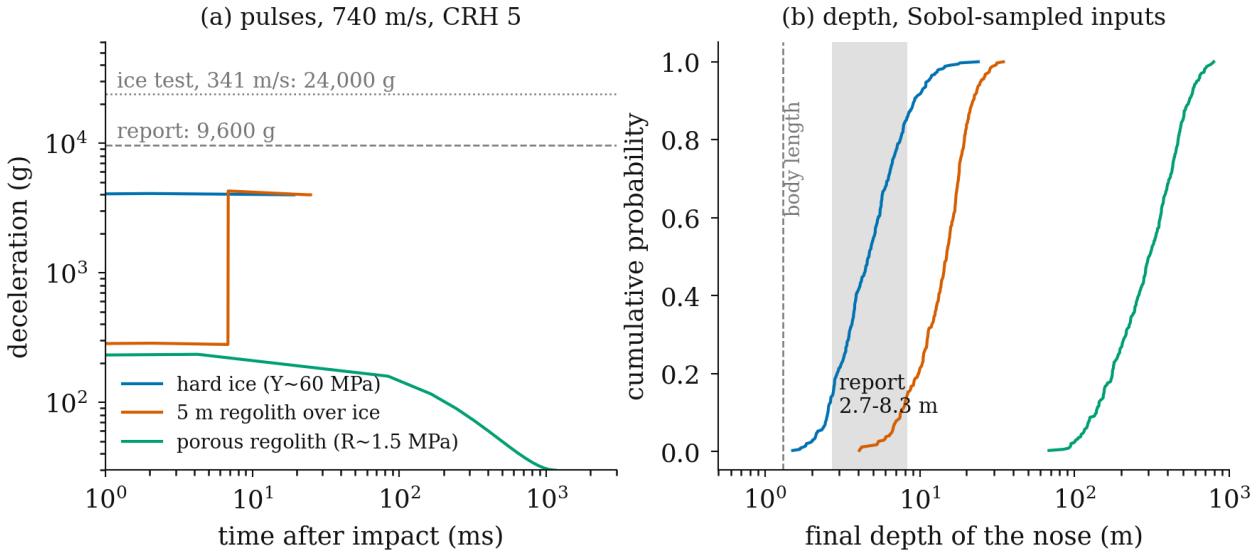


Figure 5: Left: deceleration pulses at 740 m/s for the median members of the three target classes. Right: depth distributions; the shaded band is the report's 2.7–8.3 m.

the *hard-ice* class gives a depth of 2.2–11.7 m (5–95%) and a peak deceleration of 3–18 kg. The *regolith over ice* class gives 6–24 m, and the *porous regolith* class 121–596 m at 143–357 g. Averaged over the three classes, 11% of cases exceed 10,000 g. The report's 5.4–10.4 MN and 3.4–8.3 m are therefore the stiff-ice corner of these distributions. For comparison, the empirical Sandia equation (warm ice, $S = 4.5$) predicts 39 m at 900 m/s for 119 kg with a CRH 5 nose [12].

Nose shape. In hard ice the peak deceleration is strength-dominated and nearly independent of ψ (4.4–4.1 kg from $\psi = 3$ to 12). In porous regolith it falls from 1,600 g (hemisphere) to 120 g ($\psi = 12$), while the depth rises from 70 to 435 m. A sharper nose is therefore not a free improvement, since it trades shock for burial.

Flare. A drag flare of radius 0.12–0.2 m raises the hard-ice peak deceleration from 4.2 kg to 16–42 kg and still leaves the nose 2–3 m deep, which is more than the 1.3 m body, so the antenna is buried anyway. The heritage solution is a split forebody/aftbody joined by a cable, with the antenna on the aftbody at the surface (DS-2, Mars 96).

Structure and instruments. The required ring wall (Ti-6Al-4V, 0.88 GPa yield, factor of safety 1.25, 100 kg behind the section) is 1.9/3.8/7.6/19/38 mm at 500/1,000/2,000/5,000/10,000 g. The allowable incidence for an 8 mm wall is ∼11° at 1,000 g but only 3.6° at 3,000 g (lateral load $F \sin\alpha$, $\kappa = 1$, ASSUMED). The 10° acceptance criterion of Section 5.1 is therefore generous for the hard-ice case, and it demands the 0.6° pointing given above. For instrument shock, an AC-coupled accelerometer chain with a 4 Hz corner under-reads the velocity from a 10 m pulse by 27% (5–9% after deconvolution with a ±25% corner error). The Artius broadband seismometer has no shock rating in the report, and the only demonstrated seismometer survival is 8,000–10,000 g (Lunar-A) [18].

Oblique impact or tumbling to cut peak g fails, because the normal-velocity gain is small compared with the lateral bending load at the allowable incidence (Section 5.7).

5.3 Thermal

5.3.1 What Table 34 Actually Says

The report's 70-node model is a cuboid. Its ice cases hold the battery at a constant heat load and ask when it crosses −55°C. We find that a one-node model, $C\,dT/dt = P - G(T - T_{\text{ice}})$, where C is the heat capacity, P the heater power and G the conductance to the ice, reproduces Table 34 with 3.4 W rms for $G = 0.92$ W/K and $C = 6.2$ kJ/K (battery only). The steady states (148 W at 70 K and 185 W at 30 K) give $G = 0.917$ and 0.918 W/K, which is the 1 W/K mount assumption (10 W/m^2/K × 0.1 m^2). Two assumptions therefore set the 140 W: the structure sits at ice temperature instantly, and only 2.6–6 kJ/K of thermal mass is inside the insulation.

5.3.2 Improved Model

In the improved model the core (batteries, electronics and instruments; $C = 6$–28 kJ/K) is connected through a conductance G_{ins} to a shell ($C_s = 45$ kJ/K), which loses heat by radial finite-volume conduction into crystalline ice ($k = 567/T$, $c = 7.49T + 90$ [22]) or porous regolith ($k = 0.01$–0.5 W/m/K). The conduction is radial 1-D with a geometric end-factor of 1.3 (ASSUMED). A full 2-D axisymmetric (r,z) model, the impact-heated sheath and shock-surviving insulation are not modelled (Section 6.4). Figure 6 compares the one-node fit with Table 34 and shows the heater power and the time history from the improved model.

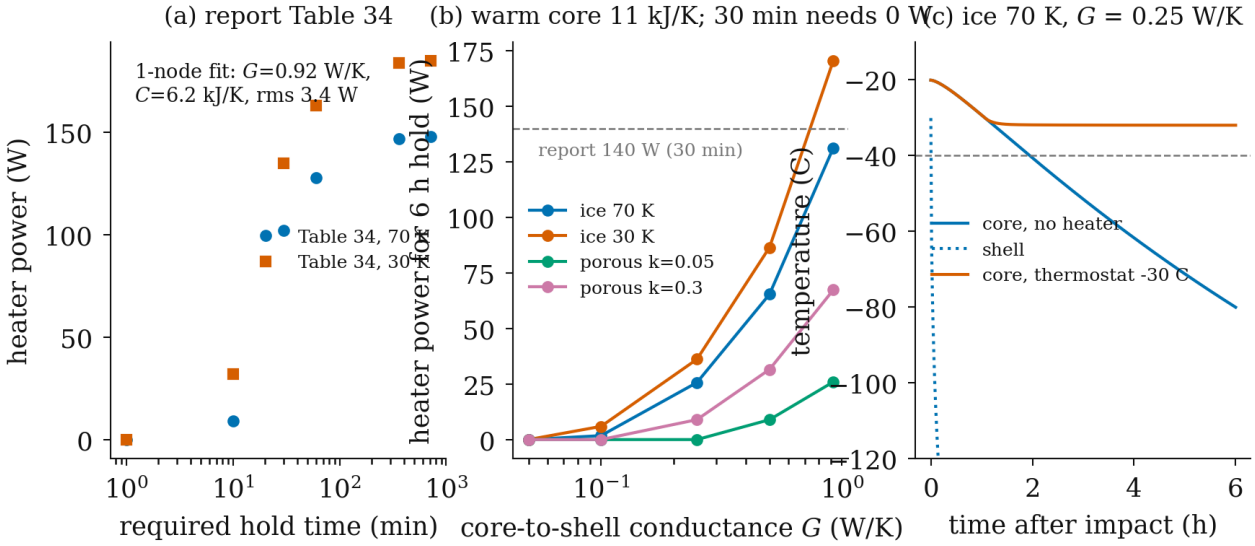


Figure 6: (a) Report Table 34 and the one-node fit. (b) Heater power for a 6 h hold versus core-to-shell conductance; 30 min needs 0 W everywhere on this plot. (c) Time history for $G = 0.25$ W/K in ice at 70 K.

5.3.3 Results

Even at the report's $G = 0.92$ W/K, a core with 11 kJ/K starting at −20°C holds −40°C for 30 min with *no* heater (0.67 h unheated at 70 K). Starting from the free-flyer equilibrium of −39°C with a −50°C limit, it needs 12.6 W. At $G = 0.25$ W/K a 2 h hold needs 11.5 W (cold start), and the unheated core survives 2.0 h in 70 K ice and 7.1 h in porous regolith ($k = 0.05$). A Sobol Monte Carlo of 512 samples (over $G = 0.08$–0.6 W/K, C, ice temperature 50–90 K, ice versus porous target, start temperature −39 to −10°C, and dissipation 3–6 W) shows that a 2 h hold needs no heater at all in 71% of cases, ≤24 W in 90%, and ≤39 W in 95%. Figure 7 shows the heater power against hold duration and the Monte Carlo distribution.

Answer to the 30-minute question. A 30-minute survival is achievable on 8 LSH20-class cells (about 1 kg) rather than 4 kg of commercial cells (Section 5.4). The *trade between heater power and duration* is shown in Figure 7: a hold of 6 h needs 0–26 W at $G \le 0.25$ and rises to 131 W at the report's $G = 0.92$. Achieving $G_{\text{ins}} \le 0.25$ W/K requires about 10 mm of aerogel-class insulation around a 0.24 m^2 core, plus Ti standoffs. The vacuum conductivity of aerogel (0.004–0.009 W/m/K) is UNVERIFIED here, and its shock survival is a TRL risk.

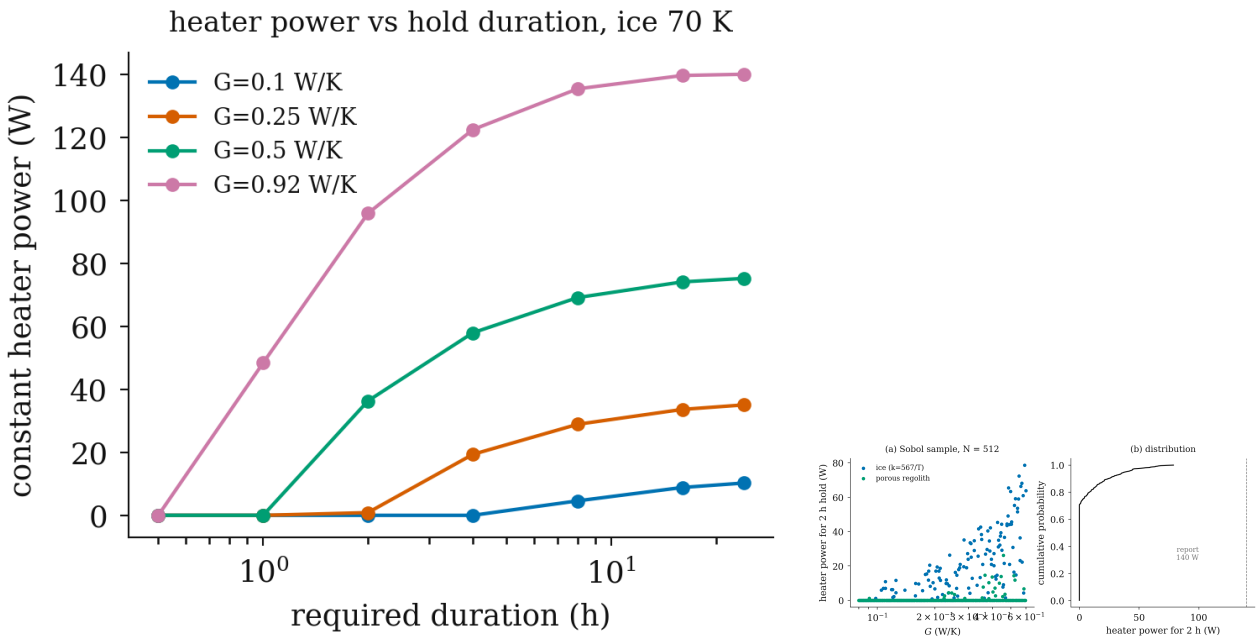


Figure 7: Left: constant heater power needed to hold the core above −40°C versus duration, for four insulation levels (ice at 70 K). Right: Monte Carlo distribution for a 2 h hold.

5.4 Power, Mass and Volume

Mass. Itemising the percentage split gives a wet CBE of 160.2 kg and a wet MEV (AIAA S-120 contingency by maturity [23]) of 191 kg, or 163 kg dry. The 180 kg limit is a dry limit (A.3), so the system fits only if the dry interpretation is confirmed, and even then only without system-level margin (the dry MEV plus 15% is 187 kg). The v2 design saves the 2–3 kg of battery and the reaction-wheel/spring flip hardware, but it stays above 180 kg wet. This is an open requirement-interpretation issue. Figure 8 shows the itemised mass and the survival time of the report's cell stack.

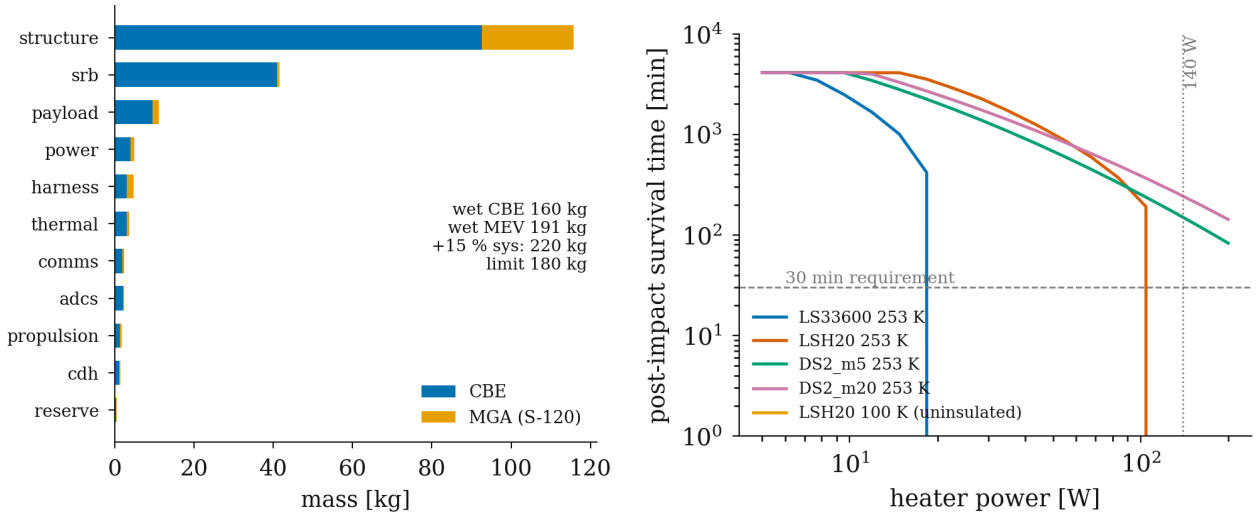


Figure 8: Left: itemised mass with contingency by maturity. Right: survival time versus heater power for a 4 kg cell stack, with cell models fitted to Saft LS33600 and LSH20 data [24, 25].

Li-SOCl$_2$ cells at cold, high current. The cell models (capacity versus temperature and current, internal resistance rising at low temperature, and voltage delay after 13.4 years) are fitted to the manufacturer plots, with an RMS error of 0.6–0.7 Ah that lies inside the ±1 Ah digitisation band. A 4 kg stack of bobbin or spiral commercial cells cannot deliver the report's 140 W at ≤ 253 K (it browns out). Only a DS-2-class cell scaled with a 5× or 20× bracket survives (150–243 min), and that result rests on a single datum [20]. The report's 4s48p topology passes 30 min at 233 K under the DS-2-class assumption. The report's total energy (195.9 Wh, against ≈1.3 kWh available in 4 kg) is not the constraint; the *pulse current at low temperature* is.

With the v2 thermal design the load is ≤20 W in a warm core, so 8 LSH20 cells (1.0 kg) suffice for 30 min at 20 W and 253 K; v2 allocates 2 kg.

v2 energy budget (ASSUMED loads). The cruise load is 0.64 W. The 4.6 h descent draws CDH 2 W, ADCS 2.1 W and comms 1 W (20% duty), for 23.5 Wh. The imagers add 10 W for the last 30 min (5 Wh), and the core preheat 10 W for the last hour (10 Wh). Post-impact, 2 h at 13.8 W (CDH, seismometer, comms and a 10 W average heater) uses 27.6 Wh. The total is ≈66 Wh, against the report's 195.9 Wh.

5.5 Communications and Data

The report's link is for a 2 GHz patch to a 1 m orbiter antenna, whereas UOP carries X-band, Ka-band and a UHF low-gain antenna [1]. We therefore rebuilt the budget at 401.585625 MHz (the Mars Proximity-1 return band [26]) with rate-1/2 convolutional coding (required E_b/N_0 of 4.4 dB, ASSUMED), a transmit gain of −3 to +3 dBi (a 0.15 m patch is only 0.2 λ at 401 MHz), an orbiter gain of 0–6 dBi, a system noise temperature of 300–800 K, and burial of 3–30 m with $\tan\delta = 1.6\times10^{-4}$–$1.6\times10^{-3}$ (plus a 5×10^{-3} tail, since no measured value at 100 K was found). Burial loss in dB per metre scales with frequency, so UHF loses five times fewer dB per metre than 2 GHz. The report's argument for 2 GHz is therefore backwards. Figure 9 shows the probability of closing the link against range and the data volume returned.

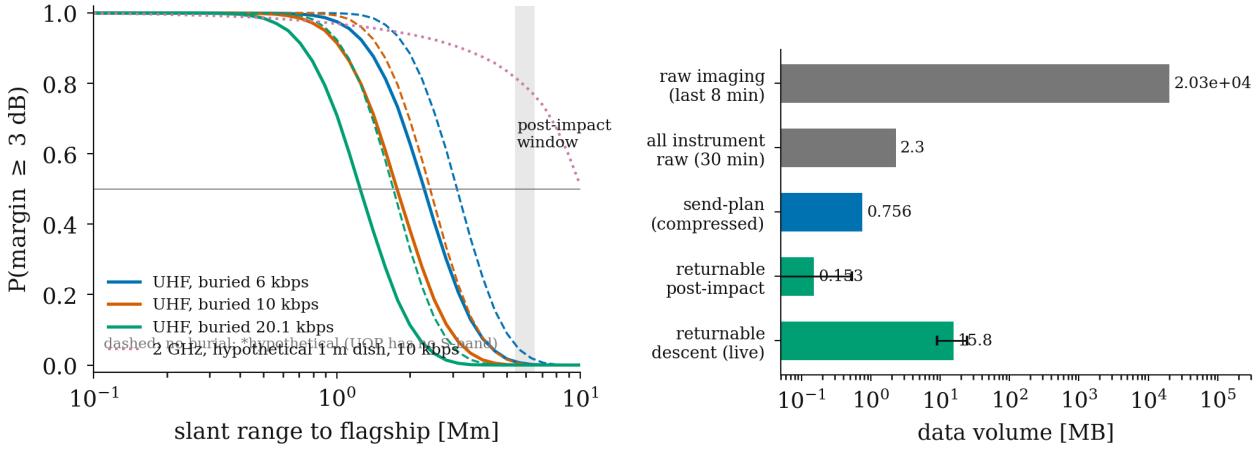


Figure 9: Left: P(margin ≥ 3 dB) versus range for the UHF link. Right: data volume returned versus data volume produced.

Results (Monte Carlo, N = 20,000, fixed seed). At 5.4–6.5 Mm the median 10 kbps margin is -7.5 dB (5–95%: −13.7 to −1.9 dB). P(margin ≥ 3 dB) is 0.4% at 6 kbps and 0% at 10 and 20.1 kbps. At 20.1 kbps the link loses its 3 dB margin at a median range of 1.7 Mm (0.9–3.0 Mm). The total data returned over the descent and 30 min is ≈16 MB (9–25 MB), of which only 0.15 MB (0.04–0.55 MB) is returned after impact. The report's 20 GB (and 50 GB memory) is 10^3–$10^5\times$ what can be returned. Reproducing "15 dB at 10 kbps at 6.5 Mm" would need an orbiter dish of 1.4 m at 2 GHz or 4.4 m at UHF (free space). *These results are dominated by the assumed antenna gains and noise temperatures. They identify the UOP UHF G/T as a required input, not a conclusion about the final system.* Choosing the second Umbriel flyby (3,653 km) shortens the range by about 1.5 Mm (+2.6 dB), at the price of a 0.89 km/s impact speed.

Store-and-forward. The data are sent in priority order (volumes in MB): accelerometer window 0.02, RTD 0.043, housekeeping 0.045, seismometer 0.65, and thumbnails 0.1 each. Only the first three fit the post-impact volume at the median.

5.6 Science Return

Nine traceability rows were built against the Decadal questions Q12.8a, Q4.4b, Q5.1c, Q5.1a and Q8.2f, each with a measurement model and an achievable uncertainty (`analysis/out/science_traceability.json`). Table 1 summarises them, and Figure 10 shows the seismic signal-to-noise ratio and the thermal-conductivity uncertainty behind the verdicts.

Table 1: Science Traceability Summary

Question	Measurement	Limit	Verdict
Q5.1c	flow stress / resistance $R(z)$	accelerometer + depth: 9% (median) to 27% (p95) on R, 13% on Y excluding model error	achievable
Q8.2f	bulk thermal conductivity	heated-cylinder transient, Cernox ±16 mK: σ_k/k 7–23% at 30 min	achievable
Q12.8a	surface imagery	only the last ∼300 s (<70 m GSD) cannot be sent live; best live image GSD ∼110 m (median)	partial
Q5.1c	porosity, coda attenuation	single station, near-surface only	partial
Q4.4b	plume/ejecta composition	plume observability from 5.4–6.5 Mm; spectrometer on UOP UNVERIFIED	not achievable
Q5.1a	interior structure, Crary waves, natural seismicity	single 30-min station; direct impact-wave range 17/46/316 km (5/50/95%); $P(\geq 1$ natural event in 30 min$)= 6\%$	not achievable

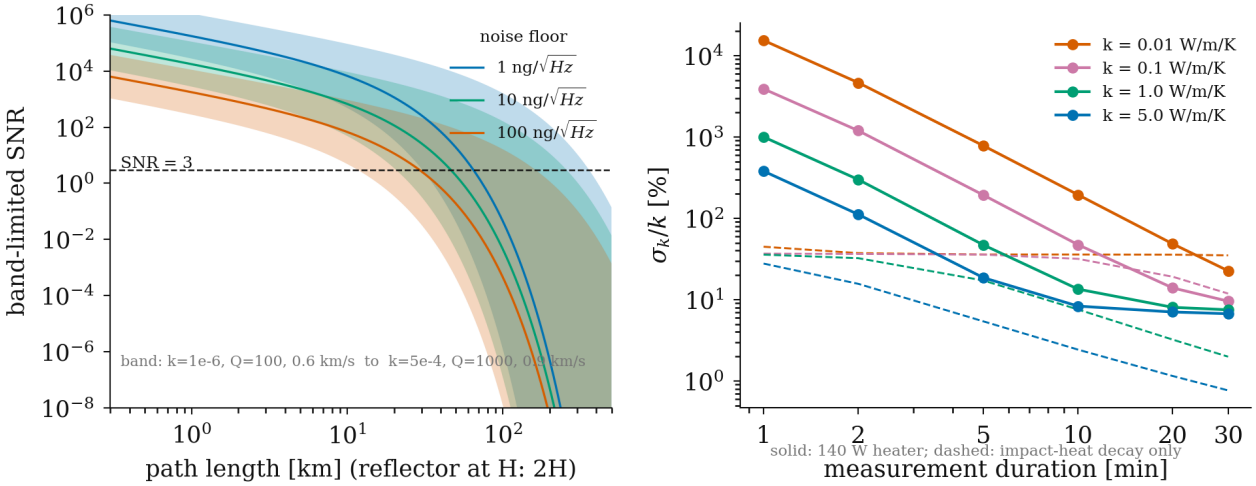


Figure 10: Left: seismic SNR of the penetrator's own impact versus distance for source efficiencies of 10^{-6}–5×10^{-4}. Right: thermal-conductivity uncertainty from the heated-body transient.

5.7 Novel-Methods Workstream

Each idea is simulated against the re-simulated report baseline (S0: 4.4 h, 401 m/s, 0.86 km/s, P(acceptable) 0.89) with the same machinery. The objectives are stated rather than hidden: minimise the impact speed v_{imp}; minimise ΔV and the propulsion mass it implies; maximise P(acceptable impact); and maximise science per kg. A strict literature check was also made. A flyby-released probe retargeted by its own burn is already in the literature (Wurz et al. 2017, with release 27 h before closest approach at Europa and ΔV of 0.5–0.9 km/s [27]; Sun et al. 2020 for Enceladus [28]; Davis 1976 for the error analysis of orbiter-launched penetrators [29]), as is the hypervelocity penetrator concept (Winglee et al., Luna-Glob, > 1.5 km/s [30]). We found no Uranus-moon penetrator study and no study specific to the UOP tour.

Figure 11 compares the release strategies, Table 2 lists the candidate designs, Table 3 gives the full scoreboard, and Table 4 the alternative target moons.

Table 2: Candidate Designs (Central Hits, Umbriel Flyby 1). Propulsion mass is SRB-scaled (S0–S2) or hydrazine-monoprop-scaled (S3–S4, Isp 220 s ASSUMED, tankage 25%, 2.5 kg hardware). The ΔV, impact speed and P ranges are over the four unknown flyby orientations.

Design	ΔV (m/s)	v_{imp} (km/s)	P(acceptable)	prop. mass (kg)	orient. tested
S0 report-like (4.4 h, 415 m/s)	401–401	0.86–0.86	0.89–0.89	33–33	1
S1 min-dV at 4.4 h (SRB)	323–328	0.75–0.76	0.94–0.94	26–26	4
S2 release 12 h (small SRB)	121–125	0.73–0.75	0.74–0.85	9–9	4
S3 release 24 h (thruster)	57–73	0.71–0.80	0.47–1.00	7–8	4
S4 release 48 h (thruster)	17–100	0.68–0.81	0.00–0.99	4–10	4

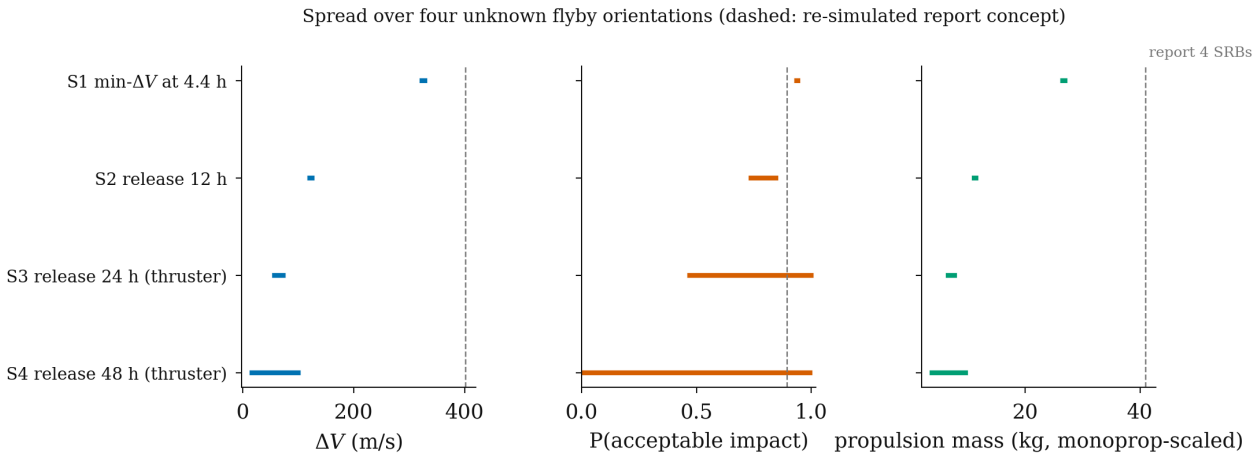


Figure 11: Spread of ΔV, P(acceptable impact) and propulsion mass over the four unknown flyby orientations for the candidate release strategies (dashed: re-simulated report concept; right-hand dashed: the report's 40.9 kg SRB stack).

What failed, and why. Burning at periapsis fails because there is no Oberth benefit at this distance. Decelerating before impact fails because, in stiff targets, the peak deceleration is set by target strength rather than speed. Oblique impact fails on bending, and a network of seismic penetrators does not fit within the mass cap. **What the numbers do not support claiming.** The numbers do not support claiming that a lower impact speed lowers shock (it does not in hard ice), that early release is robust (it is not, across flyby orientations), or that the Titania gravity-assist bound is a design.

6 Discussion

This section discusses what the results mean for the risk posture and the design, how the remaining uncertainty would be retired by test, and where the analysis itself could be wrong.

6.1 Risk Register (FMEA Recomputed)

The risk priority number is computed as RPN = $S\times$(P score $\times$ D score), following the report's convention (C.3). The P scores are binned from the model probabilities (bins $< 1, 1$–$5, 5$–$20, 20$–$50, > 50\%$). The report's cost, vendor and instrument-activation risks (CST-03 80, SHD-01 75, SCI-06 60) lie outside the models and are carried unchanged. Figure 12 compares the RPNs of the report concept and v2, and Table 5 gives the recomputed register.

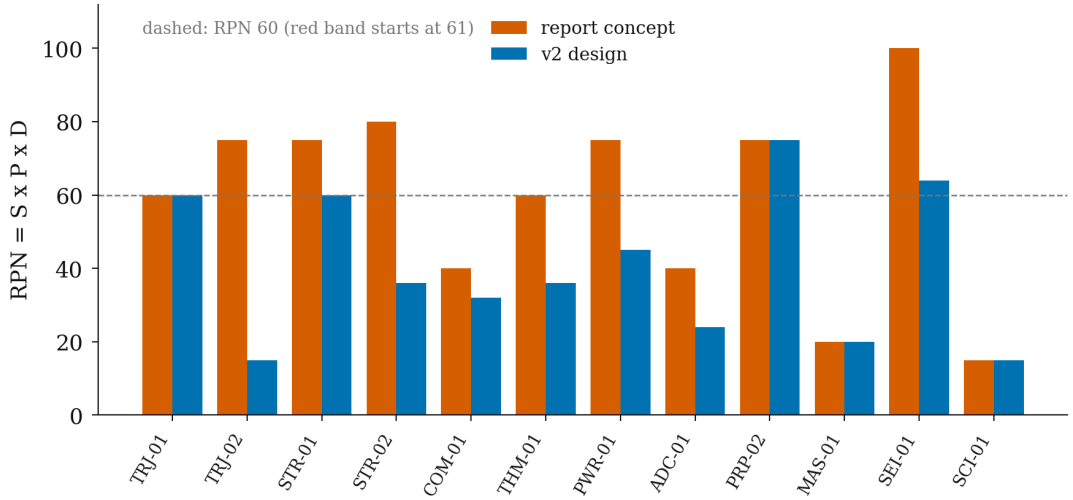


Figure 12: RPN of the report concept versus v2.

The highest residual risks after v2 are SRB ignition after 13.4 years of cold soak (RPN 75, with no verified datum), seismometer shock survival (64), peak deceleration > 10,000 g (60), and impact acceptability (60, dominated by pointing and burn repeatability).

6.2 v2 Design Summary

Table 6 lists every value changed from the report, with the reason and the source of each change.

Growth option. The Phase-A study should evaluate the 12–24 h release with a monopropellant or small SRB stage (a propulsion mass of 3–9 kg instead of 26 kg, and 5–8× the descent data), *conditional on the actual UOP tour state*, which fixes the flyby orientation. This option removes the single largest non-robust assumption (the orientation) and the largest mass line.

6.3 Verification Plan

The following tests and inputs would close the open items identified above:

1. **High-speed penetration test** (gas gun or rocket sled) of a 1/4-scale ogive penetrator into 85–100 K cold ice and ice-regolith layers at 0.5–0.9 km/s, measuring $R(z)$, depth and the g-history. This closes TRL 2 $\rightarrow$ 4.
2. **Shock qualification** of the cell stack, seismometer, accelerometer chain and insulation to 10,000 g for 1 ms, with DS-2-type cell hardware for impact (80,000 g, [20]).
3. **Cold cell characterisation**: capacity and pulse resistance of candidate Li-$SOCl_2$ cells at 233–253 K and 0.1–1 A, after accelerated 13.4-year storage (voltage delay).
4. **Thermal-vacuum** test of the warm core in a cryogenic ice/regolith simulant block, to measure G_{ins} and the heater duty directly.
5. **Link**: the UOP UHF LGA gain pattern and receiver noise temperature (a requirement request to the UOP project), and an antenna range test on the penetrator body with an ice-simulant burial.
6. **Trajectory**: the UOP tour ephemeris for the 2047 Umbriel encounter and the navigation covariance, followed by a Monte Carlo with finite-burn and spin dynamics (6-DOF).
7. **SRB**: a STAR 5D cold-soak and ignition test after thermal cycling (operating limit -55°C [21]).

6.4 Sources of Error and Model Limitations

Several simplifying assumptions in this analysis warrant discussion regarding their effect on the results.

Planar delivery model: The delivery model is planar, and the flagship's out-of-plane motion is handled only by a first-order variational equation.

Flyby orientation: The flyby orientation is unknown. The four-case spread is a bounding assumption, not a covariance.

Error budget: The error budget is ASSUMED; there are no heritage numbers for the burn repeatability, the thrust-vector offset or the Uranian-satellite ephemeris error in 2047.

Penetration model: The penetration model is a cavity-expansion model with assumed material inputs. Ice strength at 85 K and high rate has no verified datum, and no experiment exceeds 341 m/s.

Thermal model: The thermal model is radial 1-D with an end-factor. It has no impact-heated sheath, and the insulation properties are UNVERIFIED.

Link budget: The link results depend on the assumed antenna gains and noise temperature.

Seismic model: The seismic model is an energy-flux estimate, good to a factor of 3–10.

Target-class weights: The prior weights of the target classes are ASSUMED, which is why the results are given per class.

Table 7 gives our own assessment of the technology readiness of each element of the concept.

6.5 What a Reviewer Would Challenge

The objections a reviewer would most likely raise, and our responses, are as follows.

- *"Your minimum-ΔV solutions are optimal for one geometry."* Yes. The spread over four orientations is our response: the 4.4 h design is robust (P 0.94 in all four), while the early-release designs are not.

- *"Cavity-expansion models are not valid at 0.75 km/s in ice."* Agreed; no model is. The distributions bound the uncertainty, no design relies on a single class, and verification step 1 is the closing action.
- *"The thermal result depends on insulation you have not shown to survive 5,000 g."* Agreed (TRL 3). However, the unheated result is also valid at the report's own $G = 0.92$ with an 11 kJ/K core.
- *"The link numbers are pessimistic."* They are assumption-driven, and the required inputs are the UOP UHF G/T and the penetrator antenna design.
- *"Descoping Q5.1a removes the point of a seismometer."* A single co-located station cannot constrain the interior structure in 30 min (the direct range is 17–316 km against a 585 km radius), and we say so.

7 Conclusion

This work audited the 2025 PERCI concept study and rebuilt its delivery, penetration, thermal, power and link models as deterministic, tested codes with stated uncertainty, so that every number in the study could be traced and checked. The concept is not invalidated, but its headline numbers are not defensible as published.

At the subsystem level, the results show that aiming for the minimum-ΔV central hit lowers both the retarget ΔV and the impact speed, and that it also removes the need for the 180° flip. The large heater power was found to be a consequence of the thermal model rather than of the environment, and a warm, insulated core removes it for the 30-minute requirement. In contrast, the link does not close as written and only a small fraction of the planned data can be returned, so the UOP UHF receive performance becomes a required input rather than an assumption.

At the mission level, the v2 baseline meets the delivery, thermal and energy requirements with a descoped science set, while the mass fits only under the dry interpretation of the limit. Earlier release is the most promising growth option, but it is not robust to the unknown flyby orientation.

In general, the results demonstrate that a flyby-released penetrator for Umbriel is not ruled out as a secondary payload, provided its claims are stated at the strength the models support. The dominant open risk is the penetration physics at high speed into cold ice, for which no model is validated. Future work should begin with the high-speed penetration and shock-qualification tests of Section 6.3, and should repeat the delivery and link analyses once the UOP tour state and UHF G/T are known.

Acknowledgments

The author thanks the 14-person 2025 PERCI study team, whose concept study is the basis of this re-analysis; the thermal, impact, trade-study and CAD work in that study was the author's contribution. The course instructors and mentors are also thanked for their guidance during the original study.

A Audit Findings

Table 8: Audit findings (A: lead, impact/thermal/trajectory; B: independent, comms/power/ΔV/ADCS/mass/risk). Sev.: Crit. = invalidates a headline conclusion, High = changes a design number materially. Verdict: ERR confirmed error, UNS unsupported, ASM assumption, OK report is right.

ID	Sev.	PDF p.	Claim as printed	Independent recomputation	Verd.
A-01	Crit.	p.64 Table 34, p.10, p.17	Thermal: 140 W heater for 30 min	Table 34 is reproduced (rms 3.4 W) by a 1-node battery-only model with G=0.92 W/K and C=6.2 kJ/K. A realistic insulated core (11 kJ/K) at the same G needs 0 W for 30 min (12.6 W from a -39 C start). 140 W follows from the 1 W/K mount and battery-only thermal mass assumptions	ERR
A-02	High	p.55-57, Table 29	1-node analysis: 7.9 W best case	The model is a 1.6x1.06x0.96 m aluminium box (SIMPLEx envelope), not the 0.116 m-diameter penetrator (0.072 m3); the 70-node model is also a cuboid	ERR
A-03	High	p.51	Target resistance R_t = 0.6 GPa (eq B.5.3.4)	eq B.5.3.4 with sigma=0.15 GPa, E=9.1 GPa gives 0.454 GPa; 10.4 MN is reproduced only with 0.6 GPa (8.8 MN with the formula)	ERR

ID	Sev.	PDF p.	Claim as printed	Independent recomputation	Verd.
A-04	High	p.49-51	Ice flow stress 0.15 GPa from Glen's law	Glen's law is a creep law (below ~1 MPa); inversion at 0.15 GPa implies a strain rate of 4.4 /s while the text calls the ratio '4' a strain-rate ratio; A=1.3e-24 is for temperate glacier ice, not 70 K ice	ERR
A-05	High	p.50-51	Tate-Alekseevski blunt-face force for an ogive CRH-5 nose	Inertial coefficient N=0.5 (blunt) vs 0.065 (ogive CRH5): 7.7x over-prediction of the dynamic term; the report's second model (5.3 MN) is reproduced	ERR
A-06	High	p.15-16, 48-52	Impact: 'range' 5.4-10.4 MN, 3.4-8.3 m	Sobol-sampled models (3 target classes): hard ice depth 2.2-11.7 m at 2.6-18 kg; regolith over ice 6-24 m; porous regolith 120-600 m at 140-360 g. The report range is the stiff-ice corner, not an uncertainty band. The empirical Sandia/Young equation (warm ice) gives 39 m at 900 m/s	UNS
A-07	High	p.15-16, 51	No validation of any impact model at 0.75-0.9 km/s	Highest documented full-scale ice impact: 341 m/s, 24,000 g, 20 kg (S23); Young equations valid below 1219 m/s with no soft-material data at very high speed (S27). No flown penetrator has operated (S14)	ASM
A-08	Med.	p.16 vs p.52	Summary vs appendix impact numbers	6.5-16 MN and 2.7-3.5 m (Sect 6.6) vs 5.4-10.4 MN and 3.4-8.3 m (B.5.3)	ERR
A-09	High	p.15	Aft flare 'prevents over-penetration'	Flare radius 0.12-0.2 m raises hard-ice peak g from 4.2 kg to 16-42 kg and still leaves nose depth 2-3 m (more than the 1.3 m body): antenna buried. DS-2/Mars-96 use a split forebody/aftbody with a surface antenna (S10)	ERR
A-10	High	p.19 Table 2	Umbriel: 415 m/s → 901 m/s, 86 deg, 4.2 h	Independent central-hit model: 401 m/s → 862 m/s, 3.76 h (within 4% / 10%). Reproduced	OK
A-11	High	p.19 Table 2, p.18	dV 415 m/s needed; 450 m/s requirement	Minimum-dV central hit at 4.4 h is 323-328 m/s over the four unknown flyby orientations (v_imp 0.75 km/s): 22% lower dV and 13% lower impact speed than the report row	ERR
A-12	Med.	p.19 Table 2	Ariel, Oberon, Titania rows (479/414/820 m/s)	Not reproduced: Ariel 257 m/s → 0.94 km/s, Oberon 117 → 1.71 km/s, Titania(2045) 75 → 3.0 km/s. The objective used for those rows is not stated (they appear to trade dV for lower speed)	UNS
A-13	High	p.8, p.26 (SYS_SD_6)	'Titania unreachable' (820 m/s)	With the 2047-11-07 Titania flyby (0.47 km/s, 6,505 km) a central hit needs 377 m/s at 4.4 h (v_imp 0.86 km/s); the report row used the 2045 fast flyby	ERR
A-14	High	p.70-72	Retrograde burn then 180 deg flip	The minimum-dV burn direction is 67 deg from the impact velocity vector (4 orientations, 4.4 h): a slow ~67 deg slew replaces the 10 deg/s flip (see B-28)	ERR
A-15	Med.	p.14	'Crary waves' for interior structure (Q5.1a); coda from one 30-min station	Crary waves need a floating ice shell over an ocean (S38); no ocean established on Umbriel (S07). One co-located station gives near-surface Vp/attenuation only	UNS
A-16	Med.	p.12	Q4.4b composition via the flagship's 'Lucy L'Ralph' spectrometer	L'Ralph is a Lucy payload; presence on UOP UNVERIFIED (UOP payload list not confirmed); plume observation geometry at 5.4-6.5 Mm not shown	UNS
A-17	Low	p.34	Accelerometer 350B41 (-23..66 C)	An AC-coupled channel with a 4 Hz corner biases velocity from a 10 m pulse by 27%; minimum operating temperature -23 C vs core -40 C	ASM
A-18	Med.	p.57-60	70-node model: MLI e*=0.01 buried in ice	The vacuum-radiative MLI concept is not valid in compressed contact with ice; ice cases use a constant heat load, not a thermostat; no conduction into ice from first principles	ASM
B-01	Crit.	p.40-41, 10, 7	Link closes at 2 GHz to a 1 m orbiter antenna (Table 1 'S-band'; Sect 3.1.1 'UHF')	UOP lists X, Ka and a UHF LGA, no S-band. UHF 401.6 MHz at 6.5 Mm with assumed gains: 6 kbps +4..+2 dB (spread -6..+12), 20.1 kbps -1..-3 dB	ERR

ID	Sev.	PDF p.	Claim as printed	Independent recomputation	Verd.
B-02	Crit.	p.10, 15	'15 dB at 10 kbps, 1 dB at 2.14 Mbps' at ~6,500 km	Table 19 as printed: 14.3 dB (1.1 kbps), 4.8 dB (10 kbps), 1.8 dB (20.1 kbps), -18.5 dB (2.14 Mbps). The claims use EIRPs 8 dB apart; 2.14 Mbps is 106x the 20.1 kbps cap	ERR
B-03	High	p.40	RF power 5 W, EIRP 5.67 dBW	RF = 3 W (DC 5 W x 0.6); EIRP 11.4 dBW (13.66 at 5 W)	ERR
B-04	High	p.41	Rx gain 15.0 dBi (1 m, 0.6, 2 GHz)	24.2 dBi (15.0 is a 0.35 m dish)	ERR
B-05	High	p.42	B = 0.125 MHz, C/N 80.2 dB, Rmax 3.3 bps, 'Exceeds Shannon? NO', 'Above sensitivity? NO'	C/N = 20.2 dB; 80.2 dB corresponds to B = 0.125 Hz (Hz/MHz slip); Rmax = 842 kbps; the flags contradict the +30.6 dB margin	ERR
B-06	High	p.41	Ts = 70 K 'ground station at room temperature'	The receiver is on the orbiter; 290-500 K costs 6-8.5 dB	UNS
B-07	Low	p.40-41	Prx -139.1 dBW	-137.9 dBW: line and pointing losses subtracted twice (1.18 dB)	ERR
B-08	Med.	p.41-42	LoRa SF7 'modulation rate 976' with 8PSK Eb/N0 8 dB	976 is a symbol rate; LoRa is spread-spectrum so the 8PSK requirement does not apply; 1 kbps = 1024 bps	UNS
B-09	Med.	p.10	'-104 dBm at 10 deg mismatch, -118 dBm at 45 deg'	Not reproducible: the patch pointing-loss spread is 4.7 dB, not 14 dB	UNS
B-10	High	p.10	Memory 50 GB, 20 GB needed, 2.14 Mbps	The 20.1 kbps cap over 4.2 h + 30 min returns at most 42.5 MB; 20 GB would take 92 days	ERR
B-11	High	p.15, 18	In line of sight after impact because flight (4.2 h) < time before CA (4.4 h)	Closest approach is 12.5 min after impact; slant range 5.4-6.4 Mm; flagship above horizon in only 75% of impact solutions (70% at +30 min)	UNS
B-12	High	p.15-17, 41, 50-53	Flare keeps antenna at surface; 0.05-0.5 dB/m at 2 GHz	Depth exceeds body length (1.3 m); burial loss 0.1-4 dB at 2 GHz needs tan d 1.6e-4..1.6e-3 (no measurement); patch half-angle 35 deg vs flagship zenith 0-149 deg	UNS
B-13	OK	p.68	Table 37 sums and 705,145.5 J	29.175 W and 143.76 W reproduce; 705,145.5 J = 29.175 W x 4.25 h + 143.76 W x 0.5 h; 0.5936 kg and 0.7898 L reproduce	OK
B-14	Med.	p.10	'37 W avg / 140 W pulse'	37 W not reproducible (29.175 W descent, 41.2 W time-average); 140 W is a 100% duty 30 min load	ERR
B-15	Med.	p.8, 24, 11, 19, 68	Pre-impact time 4.2/4.25/4.4/4.5/4.6 h	Five different values; deploy to end of 30 min = 5.1 h, more than the 4.5 h survival requirement and the 5 h wheel life	ERR
B-16	Low	p.68, 17	330 Wh/kg and 248 Wh/L	Conservative vs LS33600 680 Wh/kg; mutually inconsistent for one 21 g, 21.6 cm3 cell; stack holds 1,331 Wh = 6.8x the need	ASM
B-17	Med.	p.17-18	4 x 2.75-3.5 V = '12 V'; 4s48p	11-14 V; capacity at -40 C 16-31% at 0.1-1.3 A; usable 224-691 Wh vs 195.9 Wh; voltage delay not analysed	ASM
B-18	Med.	p.68, 16	Heater 140 W x 0.07 duty = 9.8 W in descent	9.8x the '<1 W' survival heater; no thermal case supports it	UNS
B-19	Med.	p.13-14, 34, 68	OSIRIS-REx cameras (4 kg, 10 W each) vs Table 37 'Simera Sense' 5 W; Artius 0.6 W vs MEMS 1.5 W	Instrument table and budget disagree; two Table-13 cameras are 40 L vs a 6.8 L tube	ERR
B-20	Low	p.46, 68	ADCS 60 W x 0.035 = 2.1 W	Arithmetic OK; 60 W is per wheel; 0.035 x 4.25 h = 8.9 min vs '5 minutes'; star tracker/IMU absent	UNS
B-21	Med.	p.79 Table 46	Cell trade: DS-2 LTC wins 3.746 vs 3.6725	Totals reproduce, but a blank voltage-stability cell is scored 0 (4.444 with 0.775); margin 0.074 is smaller than the impact row 0.645; 5,000 g must-have is below the report's own 9,600 g; DS-2 cell is custom (COTS requirement); both DS-2 probes were lost	ERR
B-22	Low	p.71 Table 40	476 m/s with 4 motors	The series 139/265/379 matches base 120 kg exactly; n=4 gives 483.1 (base 119 kg: 483.3 with STAR 5D masses)	ERR
B-23	High	p.19, 71	dV margin 476 vs 450 m/s	Holds only with zero growth: 9% base-mass growth removes it; 25% growth gives 401 m/s	ASM

ID	Sev.	PDF p.	Claim as printed	Independent recomputation	Verd.
B-24	High	p.8, 26, 19	Umbriel and Titania required; Table 2 'lowest flyby velocity'	Titania row (820 m/s) unreachable with 4 SRBs, uses the 2045 fast flyby while Fig 14 lists a 0.47 km/s flyby	ERR
B-25	Low	p.70	STAR 5D data	Matches the NG catalog (10.23 kg, 3.23 empty, Isp 256 s); omitted: burn 3.03 s, 5,565 N, 13.6-19.4 g	OK
B-26	Med.	p.19, 70	SRB after 13.4 yr cruise	STAR 5D operating limit -55 C (storage -62 C); no SRB thermal case; 17 deg cant loss (4.4%) not mentioned	UNS
B-27	Med.	p.51	Impact mass 110 kg = 150-40	The budget gives 159-40 = 119 kg (149 kg at MEV)	ERR
B-28	Crit.	p.45-46, 71-72	10 deg/s SRB-ejection flip stopped by 60 mNms wheels	Needs 2.6-4.0 N m s: 43-66x the wheel; '30 mNms 5 minutes' mixes momentum and time; 1.5% impulse matching required	ERR
B-29	Low	p.71-72	Spring ejection of SRB pairs	0.5-1.2 m/s relative speed; lateral recoil 2.6-6.3 cm/s = 0.4-1.0 km miss over 4.2 h, not budgeted	ASM
B-30	Med.	p.16, 46	Spin-up 'tiny solid rocket motors'	Spin of 860-1,220 rpm needed for 1 deg with 1 mm/0.5 deg SRB offset; spin-up impulse 0.26-0.82 kN s (not tiny); Iz/It ~ 0.06 is a minor-axis spinner, unstable under dissipation	UNS
B-31	Low	p.45	Gravity gradient 1.5e-9 N m	Right order for Titania/Uranus; Umbriel's own gradient is 1.4e-8 to 1.3e-5 N m at 5,939-600 km (24 mNms integral = 40% of the wheel)	ERR
B-32	Low	p.46	Table 22 caption/units	'60mMms' (momentum unit in a torque column), informal caption, wheel life 5 h vs 5.1 h	ERR
B-33	High	p.9, 24	159 kg 'wet' of 180 kg; '12% margin'	The limit is stated dry (A.3); 25% propulsion = 4 x 10 kg so 159 is wet (dry ~131). S-120 growth of 26% predicts 193-195 kg wet (over by 13-15 kg); 21 kg = 11.7% of the limit, 13.2% of CBE	ERR
B-34	Low	p.9-10, 24, 55	Volume 0.072 m3, '>80% unused'	Consistent with 4 x 9,250 cm3 = 51.6%; the limit is quoted 0.42, 0.46, 1.19 m3 and 1.63 m3 in the thermal model	ERR
B-35	Med.	p.17-18, 34, 68	Battery 4.15 L in a 5.3 L body tube	78% of the interior; 1.4x packing exceeds it; a 20 L camera does not fit	ASM
B-36	Low	p.9	Mass budget as percentages only	14.3 kg unallocated (itemised to a 0.45 kg residual in power.py)	UNS
B-37	OK	p.80-81	RPN = S x band score	CST-03 80 = 5x16, SHD-01 75 = 5x15, SCI-06/PRP-01 60 = 5x12 are consistent	OK
B-38	Low	p.10, 80	'56 risks, 14 top risks, top three'	55 IDs visible, no rule gives 14, Table 49 lists four risks	ERR
B-39	Med.	p.80	Risk register coverage	No risk for link closure, wheel saturation, dV growth or antenna burial; goal RPNs unquantified	UNS
B-40	Med.	p.11, 19	T-4.6 h deploy / T-4.4 h burn vs Table 2	Impact is 12.5 min before closest approach on the report's timeline; the T-1 h descent is not in Table 2	ERR
B-41	Med.	p.19	Table 2 Umbriel row from an instantaneous 415 m/s burn	No direction gives 4.2 h, 901 m/s and $\geq$ 80 deg together in a 2-body check; the lead model with tides gives 862 m/s, 3.76 h (A-10)	UNS
B-42	Med.	p.15, 43	Triple 8-bit ATmegaS128 for imaging/seismic; 50 GB	No storage/processing element shown; CDH_1/2 TBR	UNS
B-43	Low	p.44	Dose 15-20 krad vs 30 krad part	Margin 1.5-2x; no Uranian-belt dose	ASM

B Reproducibility

The analysis, code and text of this successor study are kept in the PERCI_v2 workspace, and every result is deterministic. The script `analysis/run_all.py` runs every module in dependency order (trajectory sweeps, penetration, thermal, link, power, science, strategies, risk, figures and tables) and the audit, and `pytest analysis/tests` runs 52 tests (analytic limits, energy conservation, linearisation versus nonlinear integration, and determinism). All random draws use fixed seeds (link 20250611, penetration Sobol 2025, thermal Sobol 77, dispersion 11, risk 3). The parameters with provenance tags are in `research/params.json`, the sources in `research/sources.md`, and the contradictions in `research/contradictions.md`.

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Table 3: Novel-Methods Scoreboard. KEEP 4 (N1, N10, N11, N12), PARTIAL 5 (N2, N3, N5, N6, N9), UNPROVEN 1 (N13), FAILED 4 (N4, N7, N8, N14).

Idea	Result (gain over baseline, band)	Cost / caveat	Verdict
N1 Min-ΔV aim at 4.4 h	ΔV −22% (323–328 vs 401–415), v_{imp} −13% (0.75–0.76 vs 0.86), P(acc) 0.94 vs 0.89 in all 4 orientations	arrival 12 min after closest approach; needs 0.6° pointing	KEEP
N2 Earlier release (12 h)	ΔV 121–125 m/s, propulsion 9 kg vs 26–41 kg	P(acc) 0.74–0.85 (line of sight); geometry-dependent	PARTIAL
N3 Release 24–48 h with thruster	ΔV 17–100 m/s, 3–8 kg; descent data volume 5–8× (115–186 vs 22 MB scaling)	P(acc) 0.47–1.00 (24 h), 0.00–0.99 (48 h): flyby-orientation-dependent; hibernation 1–2 days at 0.64 W is cheap (15–31 Wh)	PARTIAL (best prize; Phase-A study)
N4 Burn at periapsis (Oberth)	ΔV 855 m/s at T−0.1 h vs 328 at 4.4 h; max Oberth factor 1.05	—	FAILED
N5 Staged 2+2 SRB ejection	+11.4 m/s (+2.4%) from dropping 6.5 kg of cases	extra separation event; recoil 0.4–1.0 km (B-29)	PARTIAL (not needed at v2 ΔV)
N6 Retarget aiming for low v_{imp}	v_{imp} 0.675 vs 0.756 km/s at ΔV = 390 m/s (LOS-feasible); floor 0.54	flagship range 6.9 Mm; −2.5% hard-ice peak g only	PARTIAL (small benefit)
N7 Retro motor before impact (1 STAR 5D)	−140 m/s: 0.78 → 0.64 km/s	+10.2 kg; hard-ice peak g −2.5%, depth −30%; porous g −29%	FAILED for survival (strength-dominated)
N8 Oblique impact / tumbling	—	allowable incidence 3.6° (8 mm wall, 3,000 g)	FAILED
N9 Nose shape (ψ 3–12)	hard ice: peak g 4.4 → 4.1 kg; porous: 1.6 kg → 0.12 kg, depth ×6	trades shock for burial; mass of sharper nose	PARTIAL
N10 Drag flare (report) vs split body	flare: peak g ×4–10	split forebody/aftbody (DS-2, Mars 96 heritage, not new)	KEEP split body
N11 Slow slew instead of 180° flip	wheel 0.06 N m s vs 2.6–4.0 required (43–66×); slow slew needs 0.05 N m s	450 s slew; momentum dump	KEEP
N12 Alternative target moons	see Table 4: Titania reachable (377 m/s, 0.86 km/s); Oberon 1.7 km/s (too fast); Titania-2045 3.0 km/s	Ariel 0.94 km/s; higher impact speeds raise g in porous targets only	KEEP (Umbriel flyby 1 best)
N13 Titania gravity-assist leverage	∼130 m/s equivalent per flyby, 12–14 m/s per 100 km of periapsis shift (65–71 m/s per 1,000 km)	order-of-magnitude bound, not a trajectory; needs the UOP tour ephemeris	UNPROVEN
N14 Two-penetrator seismic network	P(detect the other's impact) 0.20 at 100 km separation (2×119 kg), 0.11 (2×60 kg)	238 kg exceeds the 180 kg limit before propulsion	FAILED

Table 4: Alternative Target Flybys from the UOP Table. Configuration (+,+), release at 4.4 h and 24 h, hard-ice median target. "Prop. mass" uses the SRB-scaled rocket equation. Titania-2045 and Oberon need 75/117 m/s but arrive at 3.0/1.7 km/s.

Flyby (UOP table)	ΔV 4.4 h (m/s)	v_{imp} (km/s)	hard-ice peak g (g)	ΔV 24 h (m/s)	prop. mass 4.4 h SRB-scaled (kg)	flagship elev. +30 min (deg)
Umbriel-1 (2047-06-22)	328	0.76	4200	61	26	19
Umbriel-2 (2047-10-03)	256	0.89	4400	42	20	-1
Ariel (2047-07-29)	257	0.94	4400	50	20	3
Oberon (2047-07-28)	117	1.71	–	21	9	-56
Titania-2047-11-07	377	0.86	4300	82	31	25
Titania-2045-08-11 (report)	75	3.00	–	13	5	-77

Table 5: Recomputed Risk Register. p is the model probability; the D (detectability) scores are ASSUMED.

ID	Failure mode	S	p report	RPN concept	p v2 design	RPN	report ID
TRJ-01	Impact not acceptable (miss, incidence >10 deg, speed, flagship line of sight)	5	0.11	60	0.06	60	PRP-01
TRJ-02	Delta-V shortfall from mass growth (S-120 band)	5	0.79	75	0.00	15	PRP-01
STR-01	Peak deceleration above 10,000 g (instrument/structure shock) given target uncertainty	5	0.11	75	0.11	60	–
STR-02	Aft antenna buried (nose depth > body length 1.3 m); flare brake raises peak g 4-10x	4	1.00	80	0.15	36	–
COM-01	UHF link does not close (<3 dB margin at 6 kbps, 5.4-6.5 Mm, buried 3-30 m)	4	1.00	40	0.30	32	–
THM-01	Thermal hold for 2 h needs more than 20 W of heater (stresses cells and heater budget)	4	1.00	60	0.12	36	–
PWR-01	Cell stack brownout at cold, high current (4 kg commercial LTC at 140 W, 253 K)	5	1.00	75	0.10	45	–
ADC-01	Reaction wheel saturates in the 10 deg/s post-burn flip	4	1.00	40	0.05	24	–
PRP-02	SRB ignition failure after 13.4 yr cold soak	5	0.05	75	0.05	75	–
MAS-01	Wet mass above 180 kg at MEV (S-120 growth)	4	1.00	20	0.80	20	–
SEI-01	Broadband seismometer does not survive >1,000 g	4	0.50	100	0.40	64	–
SCI-01	Interior-structure science (Q5.1a) not achievable with one 30-min station	3	1.00	15	1.00	15	–

Table 6: Changed Values (Old → New, Reason, Source)

Quantity	Old (report)	New (v2)	Reason	Source
ΔV requirement	450 m/s (476 cap.)	330 m/s; capability 400 at MEV mass	minimum central hit 323–328 m/s over 4 orientations; capability falls with growth (B-23)	Sec. 5.1, A-11
Impact speed	0.90 km/s	0.75–0.76 km/s	min-ΔV solution	Sec. 5.1
Flight time / impact time	4.2 h, before CA	4.6 h, 12 min after CA	min-ΔV geometry	Sec. 5.1
Burn / attitude	retrograde, 10°/s 180° flip	nose-forward burn, 67° slew ≤ 0.15°/s	ΔV vector is 67° from impact velocity; wheel 43–66× short (B-28)	A-14, B-28
Impact mass	110 kg	119 CBE (149 MEV)	budget 159−40 (B-27)	B-27
Aft body	single body + drag flare	split forebody/aftbody, surface antenna	flare raises g 4–10× and leaves antenna buried	A-09
Design shock	5,400–10,400 kN, 3.4–8.3 m	distribution; design 10,000 g with isolation for seismometer/cells	Sobol result over 3 classes	Sec. 5.2
Heater	140 W (30 min)	0 W (30 min), ≤24W (2 h), preheat 10 W	core 6–28 kJ/K, $G \leq 0.25$ W/K	Sec. 5.3
Cells	192 (4 kg), 4s48p	8–12 LSH20-class, 2 kg	load ≤20 W in a warm core; cold pulse current is the limit	Sec. 5.4
Energy	195.9 Wh	≈66 Wh (ASSUMED loads)	imagers 30 min, preheat 1 h	Sec. 5.4
Comms	2 GHz, 5 W DC, 10–20.1 kbps (claim 2.14 Mbps)	UHF 401 MHz, ≤6 kbps, deployable antenna; UOP G/T required	no S-band on UOP; link does not close with assumed gains	B-01, Sec. 5.5
Memory / data	50 GB; 20 GB	∼1 GB; ≤16 MB returned	link-limited	B-10
Science	5 Decadal Q	Q5.1c, Q8.2f, partial Q12.8a	Sec. 5.6	–
Mass margin	12% (wet vs 180)	MEV 191 kg wet / 163 dry; open	S-120 growth	B-33

Table 7: Technology Readiness Assessment (Ours, Stated Honestly)

Technology (assessment, ours)	TRL	Basis
Penetration at 0.75 km/s into cold ice	2–3	highest test 341 m/s [11]; analytic only
Penetrator structure, 100+ kg, 4,000–18,000 g	3	DS-2 (3 kg, 200 m/s) lost with no cause identified [15]
Broadband seismometer at >1,000 g	3–4	Lunar-A 8,000–10,000 g prototype [18]; MEMS needs protection [19]
Li-$SOCl_2$ cells, cold, impact	4	commercial cells TRL 9 at low rate; DS-2 hardened cell custom [20]
STAR 5D motor	9 / 5	flown; 13.4 yr cold soak untested [21]
Guidance/aim from flagship, 0.6°, 20 km	3	analysis only
UHF relay link, buried antenna	3–4	Electra/Proximity-1 relay TRL 9 [26]; buried penetrator antenna not tested
Warm-core insulation under shock	3	aerogel shock survival unverified
Split forebody/aftbody	6 (at 200 m/s)	Mars 96 / DS-2 heritage [13, 14]